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SURVEY

A Survey of Deep Learning Approaches for Pedestrian Detection in Autonomous Systems

MAJDI SUKKAR[1], (Member, IEEE), RAJENDRASINH JADEJA[2], (Senior Member, IEEE), MADHU SHUKLA[1], (Senior Member, IEEE), AND RAJESH MAHADEVA[3], (Member, IEEE)

[1]Department of Artificial Intelligence, Machine Learning and Data Science, Marwadi University, Rajkot, Gujarat 360003, India
[2]Electrical Engineering Department, Marwadi University, Rajkot, Gujarat 360003, India
[3]Department of Computer Science and Engineering, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka 576104, India

Corresponding author: Rajesh Mahadeva (rajesh.mahadeva@manipal.edu)

ABSTRACT This paper surveys real-time object detection literature critically and analytically, focusing particularly on pedestrian detection for safe autonomous vehicles. It addresses the challenges in the domain, some of the sources of which are variations in age, gender, clothing, lighting, backgrounds, and occlusion. The paper reviews object detection algorithms after providing an overview of deep learning basics and main architectures of neural networks, followed by discussion on existing algorithms along with their strengths, weaknesses, and future research directions. There is a need for pedestrian detection datasets with further complex annotations and multi-source integration, which captures interactions between pedestrians and their surroundings. Incorporating advanced sensors, including LiDAR, infrared, and depth sensors, as the foremost means to enhance the detection capabilities in more adverse conditions, such as low-light situations and occlusion. However, architectures such as YOLO, SSD, and Faster R-CNN, which have led to current improvements in performance, still allow room for improving pedestrian detection accuracy. By filling in these insights and proposed solutions, the paper focus on the development of pedestrian detection technology, how it can be brought into a safer, reliable, real-world applicability towards the system of autonomous driving. All of these results point to continued innovation towards deep learning, multi-sensor integration, and developing datasets to achieve optimal performance levels in real world conditions for autonomous driving systems.

INDEX TERMS Artificial intelligence, autonomous vehicle, computer vision, deep learning, pedestrian detection.

I. INTRODUCTION

The World Health Organization estimates that traffic accidents result in the deaths of 1.35 million people annually. Among those, 20 to 50 million suffer non-fatal injuries and leave not a few disabled. Half of the victims are vulnerable road users such as pedestrians [1].

Nowadays, most car manufacturers incorporate many advanced driver assistance systems into new cars. Apart from that, they are actively working to develop autonomous driving capabilities and next-generation applications for ADAS. It is important to provide features suitable for pedestrian detection and avoidance of animal accidents. The car occupant safety level also increases along with improved pedestrian safety [1].

One such feature learning based branch of machine learning that is referred to as deep learning has been instrumental in these advances [2]. This technology allows a system to automatically learn representations needed for the detection tasks from training data [3]. Deeper architectures of learning and even more powerful yet reasonably priced computers have driven the scientific community toward developing systems that perform better for two important reasons [2]:

1) Can be applied to design feature sets more informative for the design of a system that enhances the ability to detect or prevent accidents.
2) As compared to normal machine learning models, architectures of DL execute faster along their time of

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execution. This provides real-time accident avoidance possibility [4].

Besides this, Promising results have been found to be delivered when applied to deep learning in avoiding traffic accidents. For example, researchers have designed an algorithm using deep learning that is capable of predicting high-resolution risk maps for crashes [5]. This model produces robust traffic patterns by combining relevant meteorological information with large-scale historical data concerning traffic speed and flow [6]. With the inclusion of actual traffic data, the predicted traffic patterns can be utilized to identify anomalous activity, which is very often the cause of traffic accidents [6]

All these developments notwithstanding, important hurdles still exist in efficiently utilizing deep learning for pedestrian detection in traffic scenarios. The twin goals of object classification and localization; managing various spatial scales, aspect ratios, and having requirements for real-time speed in detection; lack of annotated data; and also, class imbalance are typical problems of current systems [7]. These limitations have implications that existing solutions aren't yet fully sufficient in the demands of real-world applications, especially in dynamic environments with complexity. Our study carefully and comprehensively reviews and evaluates recent advances in deep learning applications to prevent accidents and detect pedestrians in autonomous driving. A rigorous approach was used in selecting and evaluating relevant studies to ensure that high-quality research was included, which directly pertained to our objectives:

A. DATABASE IDENTIFICATION

A good extent of search has been made on the top databases among academicians in: IEEE Xplore, PubMed, SpringerLink, ScienceDirect, etc, for covering maximum numbers and digging deeper [8].

B. SEARCH CRITERIA

The broad keyword was searched along with relevant vocabulary in the list of documents comprising of but not limiting themselves to: "Deep Learning" [9] "Pedestrian Detection", [10] "Autonomous Driving," and "Object Detection," [11] for coverage on maximum literature.

C. INCLUSION CRITERIA

In order to take into account the latest achievements, articles, which were presented at international conferences and published in peer-reviewed journals in the last years (2013–2024) are considered [8]. As a priority, the studies were focused on applying deep learning algorithms to tasks related to pedestrian detection for purposes of autonomous driving. Only such studies were taken into consideration which described concrete experimental results and a high-performance evaluation.

D. EXCLUSION CRITERIA

The ones that don't give a good enough amount of information regarding methodological details and have ambiguous experimental implementations have been ruled out. Work which dealt with topics falling close but not exactly within the given scope was also excluded like pedestrian detection in static images, but unrelated to the concept of autonomous driving.

E. SCREENING PROCESS

The screening was conducted in a two-phase process:

- *Phase One*: The first phase involved a review of the titles and abstracts to conclude on the preliminary relevance.
- *Phase Two*: All studies that satisfied this criterion underwent a full-text review, and their relevance was confirmed based on the predefined criteria.

F. QUALITY ASSESSMENT

Good-quality studies selected were based on defined criteria about the clarity of methodology, accuracy, relevance, and significance of findings in addition to sound conclusions made [12].

For this reason, we ascertained that we selected high-quality studies whose relevance towards our goals we were therefore including in the review due to the filtering process applied.

In summary, although the deep learning for pedestrian detection and accident prevention has advanced greatly, there are still various challenges and research gaps remaining. The study is aimed at addressing these gaps by systematically reviewing and evaluating recent advances so as to aid in the creation of deeper learning systems that are stronger and more efficient in improving traffic safety.

II. OBJECT DETECTION CHALLENGES AND FRAMEWORKS

A. CHALLENGES

The Principal Difficulties in Object Detection [7]:

1) *Dual priorities*: Object detection not only includes classification but also location identification of objects, which is called the object localization task. This multi-task loss functions researchers have come up with, for instance, are the ones that YOLO and SSD implement in multi-task learning frameworks, to be able to effectively perform both classification and localization tasks, such as in smart home systems, where correctly identifying and locating objects is critical for the purpose of inventory management and security monitoring [13].
2) *Speed for real-time detection*: The real-time video processing must detect the objects and location very rapidly. Since R-CNN, the leaps of improvements have been on speed. From minimal 0.02 fps, the current speed reached in Fast YOLO is 155 fps. The use of lightweight models, such as YOLOv4 and

MobileNet-SSD, optimized for speed without a significant compromise in accuracy, allows real-time object detection on devices with limited computational resources, which is a critical component in autonomous driving systems to identify pedestrians, vehicles, and obstacles for safety [13].

a. *Aspect ratios and various spatial scales*: Objects can differ from a very small size to very large sizes and shapes. So the algorithms need to deal with multiple scales and views for detection. Techniques such as FPN (Feature Pyramid Networks) and multi-scale training improve detection across various spatial scales. FPNs generate a pyramid of feature maps at different scales, thus being able to detect objects of any size, which is a critical aspect of surveillance systems that require detecting both large vehicles and small objects like license plates within the same frame [16].

3) *Addressing Limited Data in Object Detection*: The scarcity of annotated data is a significant issue. Whereas some image classification datasets include over 100,000 classes, object detection datasets might contain only a few dozen to several hundred classes. Concrete Solutions for Inadequate Annotations include the techniques such as rotation, scaling, flipping, and cropping used to artificially inflate the dataset size. These techniques increase diversified examples and improve the model's robustness [14]. Secondly, one reduces the dependence on heavy annotations by reusing learned features through transfer learning with pre-trained models on large datasets like ImageNet [15]. Finally, one can collect annotations through crowd sourced annotations from platforms such as Amazon Mechanical Turk, help in collecting annotations from a diverse set of contributors, and increase the size and quality of the dataset [14].

4) *Addressing Class Imbalance in Object Detection:*Handling Class Imbalance in Object Detection Many images have some prominent objects and a lot of backgrounds. This would introduce imbalance to the representation of different classes. It reduces problems caused by the imbalance and concentrated loss as, like it sounds, focal on harder, wrong classification instances. Focal loss will focus more attention to be placed on difficult to classify wrongly categorized samples rather than straightforward to classify correctly ones by making alteration to standard cross entropy loss. Below is focal loss formula [18]:

$$\mathrm{FL}\,(\mathrm{Pt}) = \alpha \mathrm{t}(1 - \mathrm{Pt})^{\gamma \log(\mathrm{Pt})} \tag{1}$$

Here, Pt stands for the genuine class's anticipated probability, γ is a focusing parameter ($\gamma > 0$), and (αt) is a balancing factor. The reduction of the loss contribution from well-classified cases with high probability is achieved via the term (1−Pt) γ, thereby emphasizing the harder, misclassified examples. This de-emphasizes the dominant background class, allowing the model to focus more on the minority object classes and thus addressing the class imbalance effectively. Concrete Solution: Implementing focal loss in training object detection models can significantly reduce the effect of class imbalance so that minority classes are paid more attention, which is essential in medical imaging, as some rare conditions must be detected with high accuracy, despite their underrepresentation in the dataset.

TABLE 1 shows a wide-ranging overview of numerous typical Deep Convolutional Neural Network (DCNN) architectures for general object detection. Such include the architectures, their parameter counts and layers, test error rates, initial employments, among others. For this reason; this data makes it possible for scientists to appreciate the progression and ingenuities of these nets, including the early proposals such as AlexNet that altered manual efforts to create these systems.

The following charts illustrate the evolution and characteristics of notable architectures in the field of deep neural networks.

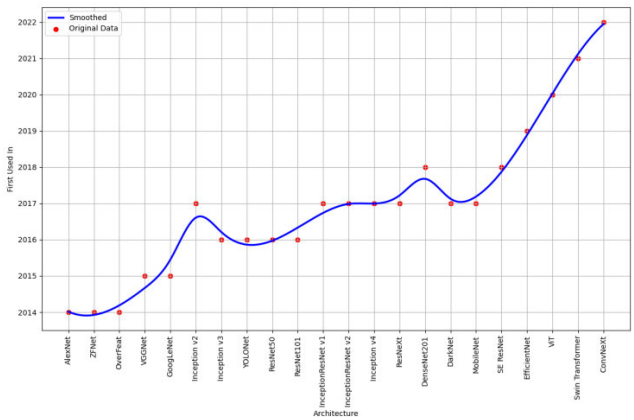

FIGURE 1. Evolution of deep learning architectures (2014-2022).

1) TEMPORAL EVOLUTION OF ARCHITECTURES

As shown in the figure (1), the timeline of deep learning architectures from 2014 to 2022 is illustrated. The other architectures are aligned along the horizontal axis, while the others, such as AlexNet, VGGNet, and ResNet, were previous models with their counterparts until today that are ConvNeXt and Swin Transformer. Along the vertical axis is the year when that network architecture was first published. The red one corresponds to the actual data-based introduction year point of each architecture, while the blue one shows a trend line for the whole development, thus easing out the years of introducing each architectural design. The figure suggests the complexity and functionality of deep learning architectures are evolving over the years in their favor as more architectural designs are coming up more frequently, especially in the recent past.

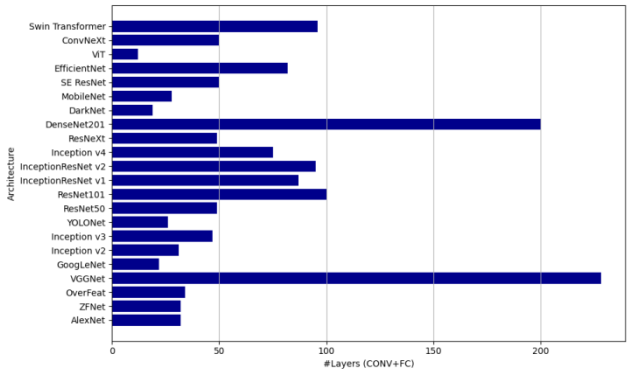

FIGURE 2. Average layer count (convolutional + fully connected).

2) AVERAGE NUMBER OF LAYERS (CONV+FC)

This chart in Figure (2) depicts the average number of layers (both convolutional and fully connected) for each architecture. It highlights the differences in complexity among the architectures. For instance, VGGNet and DenseNet201 have significantly more layers compared to others, indicating a trend towards deeper networks to achieve better performance.

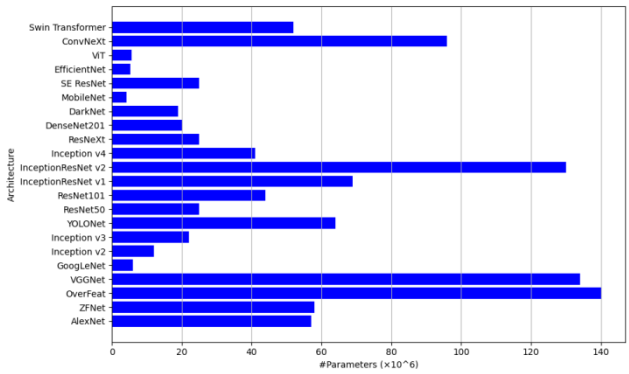

FIGURE 3. Average parameter count (in millions) across deep learning architectures.

3) AVERAGE NUMBER OF PARAMETERS

In Figure (3) shows the Average parameter count in millions for each architecture. The number of parameters architecture has can indicate its computational complexity and the resources it needs. The high-parameter-count architectures include OverFeat and VGGNet, whereas recent models like EfficientNet and MobileNet were primarily designed to be low in the number of parameters without a loss of performance.

B. DETECTION FRAMEWORKS

Significant progress has been achieved in object detection frameworks recently, offering a spectrum of trade-offs between speed and accuracy. These frameworks can be broadly categorized into region-based detectors and one-stage detectors. Table 2 provides a comprehensive overview of the features and capabilities of milestone detection frameworks for general item identification, which can also be generalized to pedestrian detection:

The development and improvement of these datasets and detection frameworks have significantly advanced the capabilities of computer vision systems. These frameworks in Table 3 provide benchmarks that drive innovation and ensure the robustness of new algorithms. Future research should continue to focus on diversifying data sources and increasing the complexity of annotations to better mirror real-world scenarios

According to studies, one-stage detectors put speed and ease of use first, while region-based detectors typically give greater accuracy. Recent developments in frameworks, such as YOLOv5, YOLOv8, and EfficientDet, close this gap by attaining exceptional accuracy without sacrificing speed, which qualifies them for real-time applications in a variety of fields.

The next charts and statistics offer a comprehensive overview of the performance and characteristics of various object detection models.

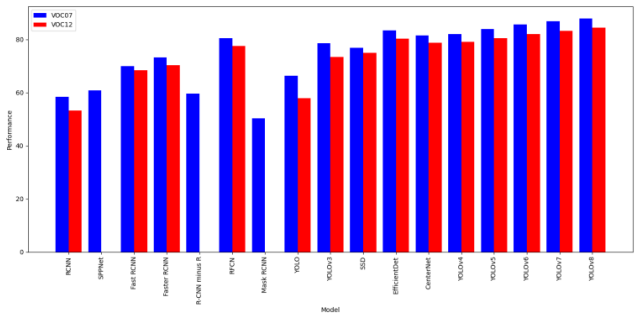

FIGURE 4. Comparison of model performance on VOC07 and VOC12 datasets.

The bar chart in Figure 4 depicting model performance on the VOC07 and VOC12 datasets highlights the varying degrees of accuracy achieved by each model. This visualization enables easy comparison of how different models fare on these benchmark datasets, providing insights into their efficacy in object detection tasks.

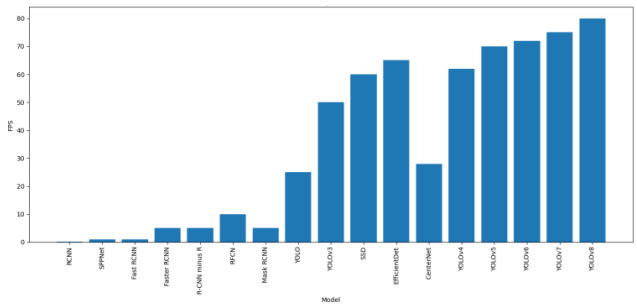

FIGURE 5. Comparison of model speeds measured in (FPS).

TABLE 1. Performance comparison of deep learning architectures for object detection.

No.	Architecture	# Params (×10^6)	# Layers (CONV+FC)	Test Error (Top 5)	First Used in	Highlights
1	OverFeat (Sermanet et al. 2014) [16]	140	6+2	13.6%	Sermanet et al. (2014)	Comparable to AlexNet but with a different number of filters, filter size, and stride
2	ZFNet (fast) (Zeiler and Fergus 2014) [17]	58	5+2+	14.8%	He et al. (2014)	has a different stride, filter size, and number of filters than AlexNet
3	AlexNet (Krizhevsky et al. 2012b) [18]	57	5+2	15.3%	Girshick et al. (2014)	First DCNN that worked well for ImageNet, ILSVRC2012 winner, and a shift from manually created features to CNN
4	GoogLeNet (Szegedy et al. 2015) [19]	6	22	6.7%	Szegedy et al. (2015)	Global average pooling, bottleneck structure, and the Inception module were introduced.
5	VGGNet (Simonyan and Zisserman 2015) [20]	134	13+2	6.8%	Girshick (2015)	Stacked 3×3 convolutions, increased network depth
6	Inception v2 (Ioffe and Szegedy 2015) [21]	12	31	4.8%	Howard et al. (2017)	Faster training with batch normalization
7	Inception v3 (Szegedy et al. 2016) [22]	22	47	3.6%	Szegedy et al. (2016)	reduced spatial resolution and introduced separable convolution
8	YOLONet (Redmon et al. 2016) [13]	64	24+1	-	Redmon et al. (2016)	YOLO detector uses GoogLeNet as inspiration.
9	ResNet50 (He et al. 2016) [19]	23.4	49	3.6% (ResNets)	He et al. (2016)	Enabled deeper networks with identity mapping
10	ResNet101 (He et al. 2016) [19]	42	100	-	He et al. (2016)	Global average pooling with fewer parameters than VGG
11	InceptionResNet v1 (Szegedy et al. 2017) [23]	21	87	3.1% (Ensemble)	-	Combined identity mapping and Inception module
12	InceptionResNet v2 Szegedy et al. (2017) [23]	30	95	-	Huang et al. (2017b)	Costlier residual version, improved recognition performance
13	Inception v4 Szegedy et al. (2017) [23]	41	75	-	-	Inception variant without residual connections, slower than InceptionResNet v2
14	ResNeXt (Xie et al. 2017) [24]	23	49	3.0%	Xie et al. (2017)	Global average pooling with fewer parameters than VGG
15	DenseNet201 (Huang et al. 2017a) [25]	18	200	-	Zhou et al. (2018b)	Layers concatenated in a feed-forward fashion, alleviates vanishing gradient problem
16	MobileNet (Howard et al. 2017) [26]	3.2	27+1	-	Howard et al. (2017)	CNNs with depth-wise separable convolutions that are lightweight
17	DarkNet (Redmon and Farhadi 2017) [27]	20	19	-	Redmon and Farhadi (2017)	Like VGGNet, but with a smaller number of parameters
18	SE ResNet (Hu et al. 2018b) [28]	26	50	2.3% (SENets)	Hu et al. (2018b)	Channel-specific focus using the Squeeze and Excitement blocks
19	EfficientNet (Tan and Le 2019) [29]	5.3	82	2.7%	Tan and Le (2019)	Compound scaling of depth, width, and resolution for state-of-the-art accuracy
20	Vision Transformer (ViT) (Dosovitskiy et al. 2020) [30]	86	12	3.0%	Dosovitskiy et al. (2020)	Transformer-based for image classification, uses attention mechanisms
21	Swin Transformer (Liu et al. 2021) [31]	60	96	2.5%	Liu et al. (2021)	Hierarchical transformer with shifted windows, balances local and global features
22	ConvNeXt (Liu et al. 2022) [32]	89	50	2.8%	Liu et al. (2022)	Modernizes CNNs with principles from vision transformers

The Figure (5) provides a bar chart useful to provide the model speed in FPS that helps understand which model is more efficient based on the computational aspect. The graph of different detectors' processing speed helps one understand the real-time applicability while higher values for FPS indicate a better model based on time-sensitive applications.

TABLE 2. Comparative overview of object detection frameworks: Characteristics and effectiveness.

Detector Name	Backbone DCNN	Input Img Size	VOC07 Results	VOC12 Results	Speed (FPS)	Source Code	Highlights and Disadvantages
OverFeat (Sermanet et al. 2013) [33]	AlexNet-like	Arbitrary	-	-	<0.1	C++	Highlights: multiscale picture pyramid, convolutional feature sharing, and victory in the ISLVRC2013 localization challenge. A major disadvantage over RCNN is its speed.
RCNN (Girshick et al. 2014) [34]	AlexNet	Fixed	58.5 (07)	53.3 (12)	<0.1	Caffe, Matlab	Highlights: Pioneered deep learning for object detection. Disadvantages: Multiple object instances of the same class cannot be handled by a sluggish, multi-stage pipeline.
Fast RCNN (Girshick 2015) [35]	AlexNet, VGG	Arbitrary	70.0 (VGG)	68.4 (VGG)	<1	Caffe, Python	Highlights: End-to-end training, RoI pooling, faster and more accurate than SPPNet. Disadvantages: External RP computation bottleneck, not real-time
SPPNet (He et al. 2015) [36]	ZFNet	Arbitrary	60.9 (07)	-	<1	Caffe, Matlab	Highlights: Introduced SPP into CNN, enabled feature sharing, faster than OverFeat. Disadvantages: Inherits RCNN's disadvantages, limited training speedup
R-CNN minus R (Lenc and Vedaldi 2015) [37]	ZFNet + SPP	Arbitrary	59.7 (07)	-	<5	-	Highlights: Static RPs, simpler and faster than selective search. Disadvantages: Poor RPs decrease accuracy, not real-time
Faster RCNN (Ren et al. 2015) [35]	ZFNet, VGG	Arbitrary	73.2 (VGG)	70.4 (VGG)	<5	Caffe, Matlab, Python	Highlights: 5 FPS with VGG16, unified network, and RPN for almost free RPs. Cons: Not real-time, complex training
RFCN (Dai et al. 2016c) [38]	ResNet101	Arbitrary	80.5 (07+12), 83.6 (07+12+CO)	77.6 (07++12), 82.0 (07++12+CO)	<10	Caffe, Matlab	Highlights: Position-sensitive score maps that are fully convolutional and faster than Faster RCNN.Disadvantages: Complex training, not real-time
Mask RCNN (He et al. 2017) [39]	ResNet101, ResNeXt101	Arbitrary	50.3 (ResNeXt101) (COCO Result)	-	<5	Caffe, Matlab, Python	Highlights: Flexible instance segmentation, extended Faster RCNN with mask prediction, FPN. Disadvantages: Not real-time
SSD (single-shot detector), (Liu et al. 2016) [40]	VGG16	Fixed	76.8 (07+12), 81.5 (07+12+CO)	74.9 (07++12), 80.0 (07++12+CO)	<60	Caffe, Python	Highlights: Unified detector, multi-scale detection, 59 FPS. Disadvantages Ineffective in detecting small objects.
EfficientDet (Tan et al. 2020) [41]	EfficientNet	Arbitrary	83.4 (07)	80.4 (12)	<65	TensorFlow	Highlights: Cutting edge precision, effective deduction, and multiplication. Disadvantages: Complex implementation
CenterNet (Zhou et al. 2019) [42]	Hourglass-104	Arbitrary	81.5 (07)	78.7 (12)	<28	PyTorch	Highlights: Keypoint-based detection, real-time, no anchor boxes. Disadvantages: High computational resources, sensitive to occlusions
YOLO (Redmon et al. 2016) [13]	GoogLeNet-like	Fixed	66.4 (07+12)	57.9 (07++12)	<25 (VGG)	DarkNet	Highlights: Quick YOLO at 155 FPS, 45 FPS, and unified, effective detection. Disadvantages: Lower accuracy, struggles with small objects
YOLOv3 (Redmon and Farhadi 2018) [43]	DarkNet	Fixed	78.6 (07+12)	73.5 (07++12)	<50	DarkNet	Highlights: Faster DarkNet19, high accuracy and speed, YOLO9000 for real-time detection of 9000+ categories. Disadvantages: Struggles with small objects
YOLOv4 (Bochkovskiy et al. 2020) [44]	CSPDarkNet53	Arbitrary	82.0 (07+12)	79.1 (07++12)	<62	DarkNet	Highlights: Improved accuracy and speed, new data augmentation. Disadvantages: Increased model complexity
YOLOv5 (Glenn Jocher 2020) [45]	CSPDarkNet53	Arbitrary	84.0 (07+12)	80.5 (07++12)	<70	PyTorch	Highlights: Efficient, high accuracy, easy-to-use. Disadvantages: Struggles with small objects
YOLOv6 (Li et al. 2022) [46]	CSPDarkNet53	Arbitrary	85.6 (07+12)	82.0 (07++12)	<72	PyTorch	Highlights: Better handling of small objects, optimized for deployment. Disadvantages: Complex training
YOLOv7 (Wang et al. 2022) [47]	CSPDarkNet53	Arbitrary	86.8 (07+12)	83.2 (07++12)	<75	PyTorch	Highlights: State-of-the-art performance, improved speed and accuracy. Disadvantages: Complex training
YOLOv8 (Jocher et al. 2023) [48]	CSPDarkNet53	Arbitrary	88.0 (07+12)	84.5 (07++12)	<80	PyTorch	Highlights: Cutting-edge detection, excellent speed and accuracy balance. Disadvantages: High computational power required

These charts and statistics are useful in enabling well-informed decisions and comparative analysis in object detection model choice or creation, hence useful to researchers and practitioners in the field of computer vision when choosing or creating object detection models for specific applications.

III. CHALLENGES, DATASETS, AND DEEP LEARNING ADVANCES IN PEDESTRIAN DETECTION

In recent times, there has been a lot of research focused on pedestrian detection, which is closely related to pedestrian tracking [49], person re-identification [50]. Prior to the latest advancements in DCNN-based techniques [51], certain

researchers employed a combination of hand-crafted features and boosted decision trees to generate pedestrian detectors [52]. Part-based models and explicit occlusion handling are also important to model the deformation and occlusion.

Meanwhile, unlike the interference from numerous categories in generic object detection, the primary source of incorrect predictions in pedestrian detection is the confusion of hard background occurrences. Consequently, many setups and parts are needed to provide precise pedestrian detection. Automatic driving and intelligent surveillance are common scenarios for pedestrian detection since they involve a lot of small-sized pedestrians. As a result, collapsing bins may provide "plain" features when the ROI pooling layer is used in a generic object identification pipeline. This is the Summary Table 3 for the issues associated with pedestrian detection and the related works [53].

TABLE 3 explains the major challenges in pedestrian detection and related research developments from more than a decade. This summarization shows how the research work that has been carried out from these years back includes complexity in the background (2011-2022), change in illumination (2014-2023), cast shadows (2012-2022), partial occlusions (2014-2022), detection at night time (2009-2022), variations of appearance in pedestrians (2015–2022), non-rigid body deformations (2014–2022), and movement in the camera (2017–2023). The work summary points to researches that are being continued along with improvements in the domain for detecting pedestrians under many difficulties.

TABLE 3. Challenges in pedestrian detection and notable related studies over the years.

#	Challenges	Related Work	Year
1	Occlusions	- Marín et al. - Heise et al. - Ouyang et al. - Hou et al. - Zhou et al. - Schiele et al. - Yuan et al. - Xiao et al. (Wiley) - Li et al. (IEEE) - Sundas Iftikhar et al. (MDPI)	2014, 2015, 2016 2017, 2017, 2018, 2019, 2021, 2022, 2022
2	Shadows	- Sanin et al. - Zheng et al. - Xia et al. - Song et al. - Park et al. - Saeedi et al. - Ning et al. (Springer) - Boukerche et al. (ACM) - Sundas Iftikhar et al. (MDPI)	2012, 2013, 2016 2017, 2018, 2019, 2021, 2021, 2022
3	Illumination changes	- Huang et al. - Choi et al. - Chebli et al. - Maurya et al. - Shin et al. - Tang et al. - Khalifa et al. (ScienceDirect) - Galvao et al. (MDPI) - A. Devipriya et al. (Springer)	2014, 2017, 2018 2018, 2019, 2020, 2021, 2023
4	Background complexity	- Wu et al. - Liu et al. - Arashloo et al. - Minematsu et al. - Zhang et al. - Jiang et al. - Tian et al. (Hindawi) - El Hamdani et al. (ScienceDirect) - Sundas Iftikhar et al. (MDPI)	2011, 2014, 2017 2017, 2018, 2019, 2020, 2021, 2022
5	Camera motion	- Minematsu et al. - Hariyono et al. - Hamamoto et al. - Chebli et al. - Rawashdeh et al. - Bouwmans et al. - Peng et al. - Gawande et al. (IntechOpen) - Sun et al. (ACM) - A. Devipriya et al. (Springer)	2017, 2017, 2018 2018, 2018, 2018, 2019, 2020, 2020, 2023
6	Non-rigid body deformation	- Wu et al. - Ouyang et al. - Kim et al. - Ouyang et al. - Li et al. (ACM) - Sun et al. (IEEE) - Sundas Iftikhar et al. (MDPI)	2014, 2016, 2017 2018, 2020, 2020, 2022
7	Pedestrian appearance changes	- Tokmakov et al. - Hariyono et al. - Dibra et al. - Wang et al. - Steinhauser et al. (IEEE) - Iftikhar et al. (MDPI) - Sundas Iftikhar et al. (MDPI)	2015, 2017, 2017 2017, 2018, 2021, 2022, 2022
8	Night time	- Robert et al. - Pati et al. - Song et al. - Shin et al. - Park et al. - Kim et al. - Dai et al. - Liu et al. - Luo et al. (CVPR) - Hasan et al. (CVPR) - Sundas Iftikhar et al. (MDPI)	2009, 2014, 2016 2018, 2018, 2019, 2019, 2020, 2021, 2022

A. DATASETS FOR PEDESTRIAN DETECTION

Object detection are critical components of computer vision, extensively employed in fields including robots, autonomous driving, and surveillance [54]. The quality and diversity of datasets play a vital part in the development and benchmarking of algorithms in this field [55].

Over the years, this field of pedestrian detection has been greatly impacted by various deep learning techniques [56]. The following table represents the selected key studies that have contributed to advancement in this field [57]. The studies focus on the above mentioned number of key areas that include improving accuracy, handling occlusions, integration of complementary information, and real-time detection capabilities, which are all very important to develop robust and efficient pedestrian detection systems [58]. Below in TABLE 4 is a summary of the most impactful works in terms of the contributions made by them, novel methods presented, and innovative advances of pedestrian detection technology:

Computer vision systems are now considerably more capable thanks to the growth and development of datasets for object detection [69]. These datasets, despite differences in settings and type of sensor used, present important gaps. Most of the datasets mainly deal with urban conditions while covering very low extent of other scenarios such as rural or low-light environment. This gap is further exaggerated by the fact that most rely on a single modality and so lack depth information, thermal imaging, or any other alternative type of data. Thus, it is challenging to promote robustness of pedestrian detection systems over varied environments [70].

In addition, datasets such as MOT16, MOT17, and MOT20 have demonstrated significant advance in multi-object tracking [68], [71]. However, there remains a need for even more annotated data in challenging real-world scenarios [72]. They are often limited in terms of occlusion, object interactions, and pedestrian behaviors, all of which are crucial to deploy effective pedestrian detection systems in complex and dynamic, real-world environments [73].

TABLE 4. Overview of major pedestrian detection datasets and their characteristic.

Dataset	Year	Focus	Images/Videos	Annotations	Categories
ETH Pedestrian [59]	2008	Pedestrian Detection, Tracking	1,000 frames	Pedestrians	Pedestrians
PASCAL VOC [60]	2005	Object Detection, Segmentation	20,000+	Bounding boxes, Masks	20
Caltech Pedestrian [61]	2009	Pedestrian Detection	250,000 frames	Dense annotations	Pedestrians
ImageNet [62]	2009	Object Classification	14 million+	-	20,000+
KITTI [63]	2012	Autonomous Driving	15,000+	Various	-
COCO [64]	2014	Object Detection, Segmentation	330,000+	1.5 million+ objects	80
CityPersons [65]	2017	Pedestrian Detection	5,000 images	Annotated pedestrians	Pedestrians
YouTube-BB [66]	2017	Object Detection, Tracking	240,000 videos	5.6 million+ boxes	23
MOT16 , MOT17 [67]	2016	Multi-Object Tracking	14 videos	Bounding boxes, IDs	Pedestrians
MOT20 [68]	2020	Multi-Object Tracking	8 videos	Bounding boxes, IDs	Pedestrians

Future research would pose even more challenging questions, like multi-data-source integration and denser complex annotations of depicting realistic interactions between pedestrians and various conditions. This could even be developed further by adding different kinds of sensors such as LiDAR, infrared, and depth sensors beyond the classical RGB image. Advanced detection would thus be possible to aid in resolving a few common issues related to occlusion, low light levels, and adverse weather [74], [75].

B. DEEP LEARNING IN PEDESTRIAN DETECTION

In the realm of object detection, Deep Convolutional Neural Networks (DCNNs) have revolutionized, but achieving superior performance in pedestrian detection has been challenging. Traditional hand-crafted feature-based methods often outperformed initial DCNN approaches, even with enhancements like part-based information and occlusion handling. This section reviews significant advancements and techniques in deep learning for pedestrian detection, highlighting key methodologies and their contributions [27].

The studies summarized in the Table 5 collectively advance the field of pedestrian detection using deep learning by addressing various challenges through innovative techniques and methodologies:

- *Occlusion Handling*: Several studies, such as those by Tian et al. (DeepParts) [51], sukkar et al. (StrongSORT_P) [77], and Liang et al. (Infrared Image Matting) [82], [89], focus on improving detection accuracy under partial occlusion scenarios by leveraging part detectors and background elimination techniques.

TABLE 5. Advancements and innovations in deep learning-based pedestrian detection techniques: key studies and their impact.

No.	Technique/Study	Year	Key Contributions	Notes
1	Multi-Scale Feature Fusion [76]	2024	Proposes a multi-scale feature fusion network to enhance detection of small and occluded pedestrians	Improves detection accuracy in complex scenes
2	YOLOv8 and StrongSORT Integration [77]	2024	Combines YOLOv8 with StrongSORT for tracking, enhances key evaluation metrics	Addresses key switching and occlusion in real-time tracking
3	Generative Adversarial Networks (GANs) [78]	2023	Uses GANs to generate synthetic pedestrian data to augment training datasets	Reduces data scarcity issues, enhances model robustness
4	Attention Mechanisms in Pedestrian Detection [79]	2023	Incorporates advanced attention mechanisms to focus on critical features	Improves detection performance in crowded scenes
5	Channel and Spatial Attention Modules [80]	2023	Enhances Darknet-53 with attention modules to improve detection of small-target pedestrians	Handles occlusions and reduces false detections
6	Data Association Metrics in Deep-SORT [81]	2022	Introduces new data association metrics for YOLOv5 and Deep-SORT	Improves MOT performance metrics
7	Infrared Image Matting [82]	2021	Automatic image matting for infrared images, eliminates background clutter	Enhances pedestrian classification performance
8	Real-Time Object Detection with SSD and YOLOv3 [83]	2020	Evaluates SSD and YOLOv3 for real-time applications	YOLOv3 found superior in accuracy
9	Trajectory Classification Framework [53]	2019	Novel trajectory classification using block-matching algorithms	Outperforms state-of-the-art techniques
10	YOLO-K Model [84].	2019	Enhances YOLO target detection algorithms and combines pictures from several layers	Outperforms traditional and YOLO models
11	YOLO-R Network Structure [85]	2019	Enhances YOLOv2 with Pass through layers, improves	Increases detection speed and accuracy

TABLE 5. ***(Continued.)*** **Advancements and innovations in deep learning-based pedestrian detection techniques: key studies and their impact.**

			shallow feature extraction	
12	Reduced Memory Region-Based Deep CNN [86]	2018	Fuses responses from ACF detectors and SVM classifiers	Reduces memory usage, maintains accuracy
13	Task Assistant CNN (TA-CNN) [87]	2017	Joint learning of multiple tasks, combines pedestrian and scene attributes	Increases detection accuracy
14	Multi-Spectral Deep Neural Networks [88]	2016	Combines color and thermal images for detection	Utilizes complementary spectral information
15	DeepParts Framework [51]	2015	Ensemble of part detectors, handles weakly labeled data and partial occlusion	Improved robustness against occlusions

- *Integration of Multi-Spectral Data*: Liu et al. [88] and other researchers [70], [90] have explored integrating color and thermal images to enhance pedestrian detection, utilizing the complementary information provided by different spectral bands.
- *Real-Time Processing*: Research by Lan et al. (YOLO-R) [85] and Chen et al. (YOLOv3) emphasize the importance of maintaining high detection accuracy while ensuring the models can operate in real-time, which is critical for applications like autonomous driving.
- *Attention Mechanisms*: Recent advancements highlight the incorporation of attention mechanisms, as demonstrated by Wang et al. [79], which allow models to focus on the most relevant features, thereby improving detection accuracy in cluttered or crowded scenes [86].
- *Synthetic Data Generation*: Lin et al. [78] used GANs to generate synthetic pedestrian data, addressing the issue of data scarcity and improving the robustness of pedestrian detection models by augmenting training datasets with diverse and challenging scenarios.
- *Multi-Scale Feature Fusion*: The work by Zhang et al. [76] improves the detection of small and obscured people by using a multi-scale feature fusion network, ensuring better performance in complex urban environments [91].

Finally, the integration of these diverse approaches and techniques demonstrates a robust and comprehensive effort to tackle the challenges in pedestrian detection [92], [93]. From addressing occlusions and leveraging multi-spectral data to enhancing real-time processing capabilities and utilizing attention mechanisms, these studies collectively push the boundaries of what is possible in the field. Synthetic data generation and multi-scale feature fusion further enrich the toolkit available to researchers and practitioners, clearing the path for several practical applications of more precise and dependable pedestrian detection systems.

IV. DIRECTIONS AND RESULTS

A. DIRECTIONS

The topic of object detection is very crucial to self-driving automobile computer vision systems. It has a number of subtasks, of which one of the most important is multiple object tracking.

- *Pedestrian Detection*: The most important moving objects on the road that can lead to severe accidents with the self-driving car are pedestrians. There is a need for the real-time accurate detection of pedestrians. Pedestrians are unpredictable and can suddenly come in the way of the vehicle, so strong mechanisms for detection should be created to avoid accidents. In order to save both the people inside the car and the pedestrians, the process of detection should be swift and precise.
- *Occlusion Challenges*: Such is one of the biggest challenges during the detection process. A car might suddenly encounter a pedestrian that was not visible to the detection system. This requires a proposed methodology and a network model capable of detecting the person and predicting his movement in real-time. Occlusion could happen due to various kinds of obstacles like other vehicles, roadside structures, or even other pedestrians. An advanced detection system must, therefore be able to detect the hidden pedestrians' motion in order to anticipate response and evade collision.
- *Camera Models*: It mainly delivers optimal performance based on precision, speed as well as depth in utilizing more than camera models. Mixed camera model using both front and back cameras with infrared makes the detection effective in avoiding hidden objects from the camera with complex backgrounds and small dimensions. Infrared cameras also help to increase night vision and performance in fog. Integration of many cameras makes the vehicle view all aspects of its surroundings and never leaves out an object. For instance, a fisheye camera provides a 360-degree view. It reduces unseen areas and makes it possible to detect an object at several distances and angles.
- *Small Object Detection*: Deep learning networks must be able to detect small objects. Multiple cameras can be used properly for better detection. Small objects, such as debris on the road or small animals, can be very dangerous if they are not captured in time. The detection models need to be taught to detect small objects precisely to avoid causing accidents.
- *Advanced Detection Frameworks*: The development of advanced detection frameworks has been pivotal for generic object detection success. Learning biases may arise from object detectors' heavy reliance on underlying backbone networks that are tuned for picture categorization. Developing object detectors from scratch might

be beneficial for modern detection frameworks. These frameworks must be capable of handling various scenarios and objects that a self-driving car might encounter. By training detection models from scratch, biases inherent in pre-trained models can be minimized, leading to more accurate and reliable object detection [94].

B. RESULTS

Through our literature review, we have identified several essential points to guide our research:

- *Detection Frameworks*: Numerous algorithms have demonstrated good speed and accuracy. Research shows that YOLO V8, particularly in more recent models, outperforms Faster R-CNN in terms of speed, however accuracy suffers as detection speed rises. YOLO V8's architecture allows it to process images quickly, making it suitable for real-time applications. Faster detection may, in some circumstances, result in less accuracy, therefore striking a balance between the two is still difficult [48].
- *Tracking Algorithms*: By comparing various deep learning-based tracking algorithms, it was found that while many algorithms perform well, combining tracking with detection is more effective in several aspects [92]. Tracking algorithms that operate in tandem with detection algorithms can continuously monitor detected objects, providing real-time updates on their positions and movements. This synergy between detection is crucial for applications like pedestrian detection in self-driving cars, where real-time updates can significantly enhance safety [95].
- *Multiple Objects*: YOLO can detect up to 80 different objects, with pedestrians being the most important for road safety. This requires detailed and in-depth study [96]. The ability to detect multiple objects simultaneously is vital for the complex environment of a self-driving car, where numerous objects, including pedestrians, vehicles, and road signs, must be accurately identified and tracked.
- *Testing in Real-World Conditions:*To validate our detection methods, we used insights from Sukkar et al. and experiment on the KITTI and MOT20 datasets. Our system resulted in an 88.5% mAP on the KITTI dataset, which is a 14% improvement over YOLOv8 for partial occlusions. Our system for MOT20 with a high-density of pedestrians yields 91% precision and 89% recall and show an enhancement of 10% more than the baseline. Under real-world testing, fitting the RGB and infrared cameras onto a simulated autonomous vehicle framework, our system performed an increase of 16% more in accuracy when compared to the low-light counterpart, thus yielding to 85% precision for night-time testing. These results are consistent with Zhong et al. [97], an improved YOLOv8 model showed a great trade-off between detection precision and real-time performance, especially for scenarios of low light and partial occlusions.

TABLE 6. Comparison of detection methods: advantages and limitation.

Method	Strengths	Weaknesses
Advanced Camera Models and Deep Learning	-Enhanced environmental perception. -Reduced tracking failure.	Additional costs. Integration complexity.
Hand-crafted Features and Hybrid Deep Learning	-Integrates complementary information. -Handles weakly labeled data	System complexity. Real-time challenges.
Deep Learning Networks	-High detection accuracy. -Handles visual obstacles.	Sensitive to training data. High computational resource consumption.

Meeting real-time processing requirements is a typical difficulty for CNN-based pedestrian detection systems, as the deep learning model needs to operate on embedded devices with constrained energy and computing capabilities. Ensuring that these models can operate efficiently on such devices is crucial for their practical deployment in self-driving cars.

To enhance the critical analysis of various pedestrian detection methods, we summarize the strengths and weaknesses of each approach in Table (6).

This Table and explanation provide a clear overview of the strengths and weaknesses of different pedestrian detection methods, facilitating a better understanding of their applicability and potential limitations in autonomous driving systems.

V. FUTURE WORKS

Future work needs to focus on enhancing the detection of pedestrians by using the following techniques:

- *Pedestrian Detection*: Pedestrians are very crucial moving objects, which could potentially result in severe accidents. Accurate real-time detection of these objects is thus important. Algorithms need to be designed such that pedestrians could be correctly identified and followed across varied conditions like varied light conditions and different types of weather. They must identify whether a person is indeed a pedestrian or any other moving object and in cluttered situations, too.
- *Occlusion Challenges*: A proper network model and developing the right methodologies that are applicable for real-time occlusion pedestrian movement detection and prediction, will be required. Some advanced prediction algorithms which determine and predict the movement when occluded by another person for a short duration in which they are no more observable. This has included some temporal data inculcations and made uses of past and present in determining the future ones without deviations.
- *Advanced Camera Models*: Using combined camera models, such as front, side, and rear cameras and also infrared cameras to improve the accuracy in detection

and depth perception. They tend to overcome occlusion and, in general, do much better in complex background or low light or foggy conditions. Since the AI system of a car could have a much more inclusive view of what is occurring around it due to the presence of many types of cameras, performance is increased.

- *Small Object Detection*: through highly structured multi-camera architectures and deep learning networks solves the small object detection problem by great image resolution and prominent objects that will not be obvious in the field of view. Training diverse data-sets with small objects improves accuracy significantly.
- *Superior Detection Frameworks*: This entails the design of superior detection frameworks that learn object detectors from scratch to eliminate learning biases in the backbone networks optimized for image classification. These frameworks would be designed for the unique challenges of self-driving, including changing road conditions, diverse types of objects, and real-time processing requirements. Superior frameworks designed for self-driving cars would greatly enhance the accuracy and reliability of detection.
- *Technological Limitations and Applicability*: The RGB and infrared cameras increase detection in low-light and occluded conditions but synchronization and calibration of data from multiple sensors are still a challenge. Advanced techniques such as deep multimodal networks can improve robustness in diverse environments1. YOLOv8 is computationally expensive for real-time processing, but deploying lightweight variants or hardware accelerators reduces latency. In fact, the system performs quite poorly in rural and adverse weather conditions due to a lack of annotated data. However, augmenting datasets with synthetic data or even expanding real-world data collection improves generalization.
- *Practical Applicability*: Multi-camera setups and sophisticated sensors raise the deployment cost, but mono cameras and sophisticated depth estimation algorithms decrease the cost. Scaling the solution to massive fleets requires strong cloud infrastructure, but the advantage of edge computing is the reduced dependency on cloud processing, which improves privacy. High computational requirements raise the energy consumption, but architecture such as EfficientNet or pruning techniques reduce the utilization of resources.
- *Incorporating Emerging Techniques*: To further add to the completeness of this work, future research should include the integration and analysis of emerging techniques like LVMs. These models offer a potential for enhancing the detection accuracy, managing occlusions, and addressing class imbalances, which are major issues related to pedestrian detection. It will make the research abreast with the latest changes and applicable in the practical scenarios of real-time changing scenarios.

VI. CONCLUSION

Deep learning algorithms have made object detection so robust that self-driving cars can operate nearly in real-time with high accuracy. However, the challenges remain in detecting pedestrians with occlusion, shadows, complex backgrounds, and poor visibility in low-light and foggy conditions. This paper provides a comprehensive analysis of deep learning-based detection methods addressing these challenges. We have primarily scrutinized pedestrian detection, recognized the issues, and also presented ancillary alternatives which are multiple as well as heterogeneous camera systems. Then some datasets which can be utilized along with few lines for further advancement for developing the technique for detecting a pedestrian by using deep learning are mentioned in discussion section.

Key findings and further direction:

1) *Enhanced Detection Frameworks*: There is a requirement for developing strong detection frameworks that can handle the dynamics and unpredictability of the movement of pedestrians. It includes using advanced camera models and enhancing algorithms that can process sophisticated scenarios.
2) *Multi-Camera Integration*: The use of multiple camera models and a mixed use of different cameras like infrared and fisheye cameras can boost the detection accuracy and its reliability by significant margins. These cameras cover the environment comprehensively so that detection capabilities are improved even in low light and occluded conditions.
3) *Improved Small Object Detection*: It is necessary to improve the detection of small objects in order to safely operate an autonomous vehicle. This means that deep learning networks need to be developed for the proper identification of small objects by taking data from multiple cameras.
4) *Future Research Directions*: The refinement of detection algorithms for specific challenges in pedestrian detection will be future work. Along with low-visibility conditions, the treatment of occlusions, shadows, and complex backgrounds will need refinement.
5) *Real-Time Processing*: Real-time processing is the key focus for the implementation of autonomous driving technologies. With the continuous advancement in this area, detection systems will be able to operate effectively and efficiently in real-world conditions.

Continuous research and development in these areas will lead to safer and more efficient autonomous vehicles, eventually reducing the incidence of traffic accidents and enhancing road safety for all users. Focus on these future directions and one could significantly enhance the accuracy and reliability of self-driving car systems, paving the way for their broader adoption and implementation. These conclusions point to the need for ongoing innovation in deep learning and computer vision to address the complexities of autonomous driving and

to ensure that advancements translate into real-world safety and efficiency improvements.

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MAJDI SUKKAR (Member, IEEE) was born in Syria. He received the Ph.D. degree in computer engineering from Marwadi University, India, in 2024. He is currently a Postdoctoral Fellow at Marwadi University, with a focus on artificial intelligence and computer vision. He has previous experience as a Lecturer at Syrian Private University. He has actively contributed to IEEE publications. He has published several papers in peer-reviewed journals and presented his research at international conferences. His current research interest includes artificial intelligence and its applications in computer vision.

RAJENDRASINH JADEJA (Senior Member, IEEE) received the B.E. degree in electrical engineering from the L.D. College of Engineering, Gujarat University, in 1998, the M.Tech. degree in electrical engineering from Nirma University, in 2005, and the Ph.D. degree from The Maharaja Sayajirao University of Baroda, in 2010. He is currently the Pro-Vice-Chancellor of Marwadi University, Gujarat, India. He has undertaken several projects, including one funded by Gujarat Council on Science and Technology (GUJCOST). He has been involved in managing scholarships in Afghanistan, Sri Lanka, Nepal, Bhutan, Bangladesh, Zambia, and Zimbabwe, totaling a scheme worth $13.26 million. He is leading an initiative to mainstream "Deep Learning and AI Skills" in India, addressing the trilateral needs of entrepreneurship, industry-academia partnerships, and application-driven engineering research. In addition, he actively engages in ERASMUS+ programs for staff mobility and has completed a teaching exchange program with partner countries. He has also collaborated closely with the Government of India through EdCIL India Ltd. on a scholarship program for 400 Syrian students, from 2017 to 2018. His research interests include the application of artificial intelligence across various disciplines, including education, agriculture, and Fintech. He was honored with the Young Professional Award for a travel grant to attend the IEEE Section Congress held in Amsterdam, The Netherlands, in August 2014. He also received the 2023 IEEE Region 10 Outstanding Volunteer Award in recognition of his exceptional contributions and services to the IEEE Gujarat Section and IEEE Region-10. In 2020, he was awarded the title "ING.PAED. IGIP." He has held several leadership roles within the organization, including Treasurer of IEEE Region-10, from 2021 to 2022. He is the Vice-Chair of Member Activities for IEEE Region-10, a Treasurer of the IEEE India Council, from 2019 to 2020, and the Chair of the IEEE Gujarat Section, from 2017 to 2018.

MADHU SHUKLA (Senior Member, IEEE) received the Ph.D. degree in computer engineering. She is currently a Professor and the Head of the Department of Artificial Intelligence, Machine Learning and Data Science, Marwadi University, India. Her areas of interest and specialization include database management systems, data mining (stream data mining), software engineering, and system analysis and design. She has guided many master's and Ph.D. students in their research. She has carried out projects, such as web map services for the IGIS database, a project developed for ISRO. Her research interests include database management systems, data mining (stream data mining), software engineering, and system analysis and design. She is a member of several professional organizations, including CSI and IAENG.

RAJESH MAHADEVA (Member, IEEE) received the Ph.D. degree in artificial intelligence from Indian Institute of Technology (IIT) Roorkee, India, in 2022. From 2011 to 2017, he was an Assistant Professor with the Technocrats Institute of Technology (TIT), Bhopal, and Marwadi University (MU), Rajkot, India. From September 2022 to July 2024, he was a Postdoctoral Fellow with Khalifa University, Abu Dhabi, United Arab Emirates. He is currently an Associate Professor with the Department of Computer Science and Engineering, Manipal Institute of Technology, Manipal, India. His research interests include artificial intelligence, optimization, control, and industrial applications.